

TWELVE UNIT CUBES IN A CUBE OF SIDE 2.93152

Yohei Nakajima
yohei@untapped.vc

Abstract

We give a packing of 12 non-overlapping unit cubes in a cube of side 2.9315185094797, with every pair of cubes and every wall at least 10^{-6} apart. The previous best known side was 2.9327717687... (H. Lin, July 2026). The packing was found by compressing unit-diameter balls in a shrinking box and hardening them into cubes under continued inward pressure. Coordinates, a short verifier and an exact certificate are public.

1. The problem

Let $s(n)$ be the side of the smallest cube that contains n non-overlapping unit cubes, which may be rotated. The known values are few. Friedman's catalogue [1] lists non-trivial packings only for $n = 9$ –14 and 28–33; for every other $n < 34$ the trivial packing in a $\lceil n^{1/3} \rceil$ -cube is the best known. From 1998 the best known side for $n = 12$ was $2 + 2\sqrt{2}/3 \approx 2.94281$ (Friedman); in July 2026 H. Lin improved it to 2.93277177..., with coordinates published in [2].

Theorem. *Twelve unit cubes fit in a cube of side 2.9315185094797. Hence $s(12) \leq 2.93152$.*

2. The packing

Table 1 lists the twelve cubes. Cube i is the unit cube $c_i + R(q_i) [-\frac{1}{2}, \frac{1}{2}]^3$, where c_i is its centre and $R(q_i)$ the rotation given by the unit quaternion $q_i = (w, x, y, z)$; the container is $[0, s]^3$ with $s = 2.9315185094797$. Four cubes are exactly axis-parallel, two are tilted by less than 1° , and six are tilted by 9° or more (Figure 1). Full-precision values are in the file `cubincub.n12.json` [3].

i	x	y	z	w	q_x	q_y	q_z
1	1.5243900087	2.4315173049	1.3320181026	0.5838886869	-0.5838886869	-0.3988408220	-0.3988408220
2	2.4315175095	2.4315173049	0.5000010000	0.7071067812	0.7071067812	-0.0000000000	0.0000000000
3	0.5000010000	1.4369556928	1.4377218077	0.5395289954	0.4570650535	0.4570650535	-0.5395289954
4	0.5098745912	0.5769319953	2.3545775829	0.4560511351	-0.5403863020	-0.4560472394	0.5403896022
5	0.6086104699	2.2929309153	2.3887769910	0.0258332833	0.7609821302	-0.6482073268	-0.0081302199
6	2.4315175095	1.4315163049	0.5000010000	0.0000000000	-0.0000000000	-0.0000000000	-1.0000000000
7	2.4298431049	2.4215276465	2.2620542416	0.0000269579	0.0016621483	-0.9999712786	0.0073944948
8	1.5800409188	1.4806927085	2.4315175095	0.0000000000	-0.9765112947	-0.2154662186	-0.0000000000
9	1.6870691741	0.7046058868	1.3931640729	0.1630483719	-0.8266039691	-0.3023204668	0.4458065074
10	0.6714203714	0.5125513149	0.5042794969	0.0019279356	0.0017767559	-0.7100527881	-0.7041435680
11	2.4315175095	0.5648571561	2.4315175095	0.5000000000	-0.5000000000	-0.5000000000	0.5000000000
12	0.5000010000	2.3777515563	0.5000010000	0.7071067812	-0.7071067812	0.0000000000	0.0000000000

Table 1. Centres and rotation quaternions, rounded to 10 decimals.

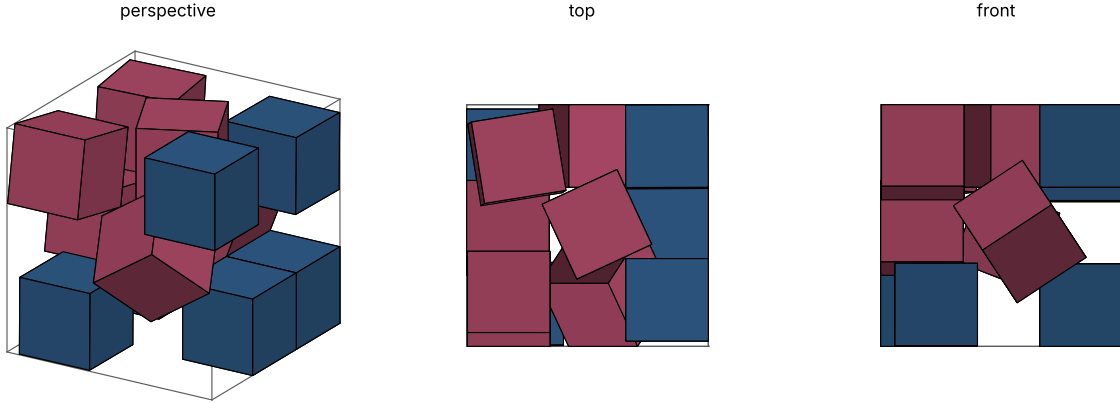


Figure 1. The packing from three directions. Red cubes are tilted.

3. Verification

For each cube, rotate the eight corners $(\pm\frac{1}{2}, \pm\frac{1}{2}, \pm\frac{1}{2})$ by q_i and translate by c_i . All corners lie in $[0, s]^3$; the smallest distance from a corner to a wall is $1.000000000 \cdot 10^{-6}$. Two boxes are disjoint if and only if one of fifteen axes separates their projections: the three face normals of each cube and the nine cross products of their edge directions. Over all 66 pairs the smallest separation along such an axis is $1.000000001 \cdot 10^{-6}$. A 30-line program that performs these checks, `verify.py`, is in [3].

The same checks were repeated in exact rational arithmetic. Every number in the file was converted to a rational, each rotation was taken as $H(q)/|q|^2$ (exactly orthogonal for any $q \neq 0$, where H is the homogeneous quaternion matrix), every corner was confirmed to lie in $[0, s]^3$, and for each pair an explicit rational plane was exhibited that strictly separates the two cubes. The program is `certify_exact.py` in [3].

4. How it was found

Each cube begins as a ball of diameter 1 and is replaced, over the course of the search, by the *rounded cube* of softness r : a cube of half-side $\frac{1}{2} - r$ enlarged by a ball of radius r . At $r = \frac{1}{2}$ this is the ball, at $r = 0$ the unit cube, and every intermediate shape lies inside the unit cube. Starting from random positions, the balls are compressed under inward pressure on the container; then r is lowered to 0 while the pressure continues, so the shapes grow into cubes as the box settles around them. Overlaps are penalized and positions, orientations and the box size follow gradient descent. The best configuration was then tightened by a constrained optimizer that keeps every corner inside the box and a separating plane between every nearby pair, and finally spread by a factor $1 + 10^{-6}$ to give the stated clearance.

Two of 192 random starts landed below the previous record, in two different arrangements (2.931515 and 2.931620 before clearance was added). With rigid cubes from the start, the same schedule did worse: on seeds 1–32, one rigid start and 13 ball starts ended below the trivial side 3, and the best of 32 rigid starts was 2.99257. On squares in a square, the method recovers the

best known packings for $n = 11, 17, 18, 19, 26$ and 27 from random starts [3].

5. Remarks

Several quaternions in Table 1 have repeated components, which suggests a closed form for the side. We have not found it. We use no lower bound beyond the volume bound $s(12) \geq 12^{1/3} \approx 2.289$; how close 2.93152 is to optimal is open. For $n = 11$ the same search reached $2 + 2/\sqrt{5} \approx 2.894427$, above Lin's 2.88295; for $n = 13$ and 14 it did not improve the catalogue.

Acknowledgement

Code, experiments and verification were developed with Claude (Anthropic), an AI system, under the author's direction. The author takes responsibility for the content.

References

- [1] E. Friedman, *Cubes in Cubes*, Erich's Packing Center, <https://erich-friedman.github.io/packing/cubincub/> (accessed 8 October 2026).
- [2] Tencent Hunyuan, *Hyra-results: packing records*, file `cubincub_n12.json`, <https://github.com/Tencent-Hunyuan/Hyra-results> (accessed 8 October 2026).
- [3] Y. Nakajima, *soft-to-rigid-packing*, version 1.0.0, <https://github.com/yoheinakajima/soft-to-rigid-packing> (2026), doi:10.5281/zenodo.23248095.